

A Benchmark of Eye Anterior Image Semantic Segmentation in Scarcely Labeled Datasets

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Abstract—Semantic segmentation of anterior-segment ophthalmic images is hampered by scarce pixel-wise annotations. We benchmark three data-efficient training paradigms built on a foundation model: (i) parameter-efficient fine-tuning of *Segment Anything* (SAM) via LoRA, (ii) weakly supervised learning that converts Grad-CAM activation maps from a ResNet-18 classifier into SAM-prompted pseudo-masks, and (iii) few-shot segmentation using feature-density prototypes. Experiments on the public pupillary segmentation Dataset (299 images) evaluate Intersection over Union (IoU), averaging results over 10 runs per configuration. LoRA achieves the best and most stable performance (mean IoU ≈ 0.97 across $N=42$ –210 labeled masks). The weakly supervised pipeline attains competitive accuracy without pixel-wise labels (peak IoU 0.85 $N=168$) but is sensitive to thresholding. The few-shot method reaches IoU 0.75 with $k=20$ masks and offers a viable path when only a handful of annotations are available, albeit with higher variance. These findings delineate a practical “supervision ladder”: use LoRA when tens of masks can be curated; prefer weak supervision when only image-level labels exist; and resort to few-shot learning for minimal mask budgets. Overall, coupling foundation models with annotation-efficient training enables clinically useful ophthalmic segmentation under realistic data scarcity.

Keywords—Ophthalmology, Anterior-Segment Imaging, pupillary segmentation, Semantic Segmentation, Segment Anything Model, LoRA/QLoRA, Weakly Supervised Learning, Grad-CAM, Few-Shot Learning

I. INTRODUCTION

Deep learning has transformed medical image segmentation, enabling high-accuracy identification of ocular lesions associated with glaucoma [1], [2], diabetic retinopathy [3], cataract [4], Ocular Hypertension [5], and age-related macular degeneration [6], [7]. This capability supports earlier diagnosis and more precise clinical monitoring.

Concurrently, foundation segmentation models have emerged, notably the Segment Anything family (Segment Anything Model (SAM) [8], SAM2 [9]) and specialized medical variants (MedSAM [10], MedSAM2 [11], FastSAM [12], MobileSAM [13]). Nevertheless, their training and adaptation typically rely on massive corpora that far exceed the scale of open ophthalmic repositories; in our case, the working dataset comprises 299 non-augmented images. Practical barriers in clinical settings, expert annotation cost and time, volumetric studies, and stringent data-privacy regulations further hinder the construction of large, richly annotated datasets [14]–[17].

This landscape motivates low-annotation strategies that achieve competitive performance with limited resources: few-shot learning [18], semi-supervised learning [19]–[23], and weakly supervised learning via pseudo-labeling [24]. Complementarily, lightweight

adapters such as Low Rank Adaptation (LoRA)/QLoRA enable the customization of large models without requiring massive annotation. Taken together, these approaches substantially reduce the labeling burden without materially compromising result quality.

In this article, we compare three low-annotation strategies for ophthalmic image segmentation: (i) few-shot learning based on feature densities, (ii) weakly supervised learning, and (iii) LoRA-based fine-tuning of the SAM. We evaluate these approaches under multiple small-data regimes by varying the number of training images. While LoRA fine-tuning on SAM attains the strongest overall performance, our analysis highlights the practicality of techniques that operate with very limited data, an essential advantage in ophthalmology, where labeled examples are scarce.

A. Problem Definition

This study evaluates the effectiveness of three specific low-data training techniques: *Few-Shot Learning (FSL) using Feature Densities (FD)*, *Weakly Supervised Semantic Segmentation (WSSS) using Gradient-weighted Class Activation Mapping (Grad-CAM)s*, and *Finetuning using LoRA*. To this end, one advanced segmentation models, SAM, is employed, focusing on one ophthalmic task characterized by limited data availability: pupillary segmentation (299 images). Our primary objective is to assess how these low-data training techniques affect model generalization, thereby identifying optimal methodologies for implementation in clinical environments with limited resources.

B. Contributions

We tackle pupillary segmentation in anterior-segment images under three paradigms (Few-Shot, WSSS, and LoRA-based fine-tuning), with a pipeline directly applicable to small clinical datasets and limited annotation budgets.

- 1) **Few-Shot:** we leverage *feature densities* over ResNet-18 embeddings and candidate masks produced by SAM [8].
- 2) **WSSS:** we use Grad-CAM [25] as a localization prior to guide SAM [8], selecting the final mask by consistency with that guidance.
- 3) **LoRA fine-tuning:** we adapt SAM by training low-rank adapters [26] while freezing most parameters.
- 4) **Evaluation:** we assess on the public *pupillary segmentation Dataset* [27], using pixel-wise labels strictly for evaluation; we benchmark against a LoRA-tuned SAM baseline and report Intersection Over the Union (IoU) averaged across configurations (10 runs per configuration).

II. STATE OF THE ART

A. Ophthalmology resources for detecting cataract with AI

Cataracts remain a leading cause of blindness, yet research output lags behind clinical need: according to [28], over sixty years (1965–2025) only 12 original studies proposed diagnostic methods

for low-resource settings, and just 8 incorporated Artificial Intelligence (AI)-based tools, typically reporting accuracies above 90%. This highlights a substantial gap between the condition's clinical importance and the amount of dedicated research, particularly where resource constraints limit access to specialized ophthalmic care.

Recent AI-based studies show promising, practice-oriented results, including a mobile application achieving $\sim 90\%$ performance for cataract screening in resource-limited environments [29]. Moreover, Convolutional Neural Network (CNN)-based approaches—computationally efficient and interpretable—deliver competitive performance [30]; thus, well-designed pipelines that balance model capacity with deployment constraints should be prioritized, leveraging data-efficient training paradigms to cope with limited annotations.

B. Data-poor training paradigms

1) *Semi-supervised learning*: Semi-Supervised Learning (SSL) is a key advance for medical image segmentation, where high-quality labels are scarce but unlabeled images are abundant [31]. By exploiting both sources, SSL improves performance without extensive manual annotation.

Progress in the past four years has been rapid [32]. Key milestones include: **Dual-Task Consistency** (2021), coupling pixel-wise segmentation with level-set geometry via a cross-task consistency loss for pancreas/left-atrium data [33]; **Cross Teaching (CNN-Transformer)** (2022), mutual pseudo-labeling that leverages architectural diversity and outperforms ACDC baselines without explicit consistency [34]; **DDFP** (2024), feature-level consistency guided by density estimation, reaching state of the art on Pascal VOC/Cityscapes [35]; **AllSpark** (2024), re-generating labeled features from unlabeled ones with channel-wise cross-attention and semantic memory [36]; **ABD** (2024), bidirectional perturbations for labeled/unlabeled data that encourage consistency and exploration [37]; **SynFoC** (2025), seeding training with strong pseudo-labels from a foundation model (e.g., MedSAM) and refining with a U-Net across domains [38]; **SemiDAViL** (2025), blending SSL, domain adaptation, and vision-language priors via DLG and DyCE [39]; and **AmbiSSL** (2025), addressing inter-annotator variability with diverse pseudo-labels, latent distribution learning, and cross-decoder supervision [40].

Overall, SSL is shifting from hand-crafted consistency toward versatile, multi-modal systems that fuse visual, geometric, and vision-language cues while tackling domain shift and annotation noise. Two hurdles remain: principled selection/weighting of unlabeled data under distribution drift, and understanding how pseudo-labels propagate uncertainty through deep decoders.

2) *Few-shot learning for medical image segmentation*: FSL has emerged as a practical answer to the annotation bottleneck in medical image segmentation: models must generalize from only a handful of labeled examples while maintaining clinical fidelity. Recent surveys consolidate the field and highlight meta-learning, prototype-based matching, and parameter-efficient adaptation of foundation models as the dominant families of approaches [18], [41]. Below, we summarize representative advances from the last few years, with emphasis on methods relevant to ophthalmic imaging when available. **Global Correlation Matching** (2022). Introduced a global correlation module to better align support and query features in episodic training, improving generalization across abdominal CT/MRI classes under extreme data scarcity [42]. **Self-supervised pretraining for FSL** (2022). Showed that self-supervised objectives can replace fully labeled base classes during meta-training, yielding competitive Dice scores across cardiac and abdominal tasks using superpixel-derived pseudo-labels [43]. **Multi-prototype descriptors** (2024). Proposed generating multiple representative descriptors to mitigate prototype bias and capture intra-class variability, reporting

consistent gains on multiple benchmarks [44]. **Ophthalmology, one-shot AV segmentation** (2023). Leveraged cross-modality pretraining to realize *one-shot* retinal artery/vein segmentation across diverse datasets, a concrete demonstration of FSL viability in retinal analysis [45]. **Foundation models with domain tuning** (2024/2025). Parameter-efficient adaptation of SAM-like encoders has been explored to reduce the number of learnable parameters and labeled images needed for specialization. A representative example is DT-SAM, which inserts lightweight adapters into the SAM encoder and couples them with a U-Net-style decoder for effective few-shot transfer [46].

3) *Weakly supervised learning for medical image segmentation*: WSSS reduces the reliance on dense pixel annotations by training with sparse or coarse labels (e.g., image-level, scribbles, or bounding boxes). In medical imaging, and ophthalmology specifically, WSSS has proven valuable where annotation budgets are limited or expert time is constrained.

Scribble supervision: dual-branch and pseudo-labeling (2022). Proposed a dual-branch network that dynamically mixes predictions to form reliable pseudo-labels from scribbles, outperforming prior scribble-based and several semi-supervised baselines on ACDC [47]. In a clinical application, scribble-supervised meibography achieved competitive gland segmentation with substantial annotation savings [48]. **Box supervision and MIL extensions** (2023). To relax spatial supervision further, combined loose bounding boxes with a detect-grow-segment strategy, showing that careful MIL design can close much of the gap to fully supervised training [49]. **Image-level supervision in retinal OCT** (2024). For ophthalmic lesions, Yang *et al.* introduced an anomaly-guided mechanism that uses image-level labels and anomaly cues to drive pseudo-label refinement, achieving state-of-the-art weakly supervised OCT segmentation [50]. **General scribble pipelines and transformers** (2022–2024). Cycle-consistency and mix-based augmentation (CycleMix) improved robustness to sparse supervision [51], while more recent CNN-Transformer hybrids (e.g., ScribFormer) enhanced long-range context modeling under scribble constraints [52]. **Toward retinal layer parsing with weak labels** (2025). For OCT layer segmentation, central to many retinal workflows, weakly supervised strategies with uncertainty-aware prototypes and boundary regression have reduced dependence on dense labels while preserving anatomical fidelity [53].

C. Segmentation models overview

a) *Classic Segmentation Models*: Models based on convolutional architectures have been fundamental in the development of semantic segmentation, particularly in the medical domain. Among the most representative are UNET [54], DeepLabv3+ [55], and Mask R-CNN [56], widely adopted for their effectiveness and adaptability.

UNET proposed by [54], is a U-shaped architecture composed of a symmetric encoder-decoder structure, connected through skip connections that merge feature maps at different resolutions [57]. Its design enables accurate training even with limited datasets, making it especially suitable for medical applications. Building on this foundation, **DeepLabv3+** [55] builds upon **DeepLabv3** [58] by incorporating a decoder module that enhances segmentation along object boundaries. The architecture retains an encoder based on atrous convolutions to capture multi-scale contextual information, while the decoder effectively refines segmented contours. In parallel, **Mask R-CNN** [56] extends Faster R-CNN by adding an additional branch that predicts pixel-wise segmentation masks for each Region of Interest (RoI). This branch, implemented as a fully convolutional network, operates in parallel with classification and bounding box regression, and is essential for producing precise, localized segmentations. These models have established a robust foundation upon which more recent and generalist segmentation approaches have been developed.

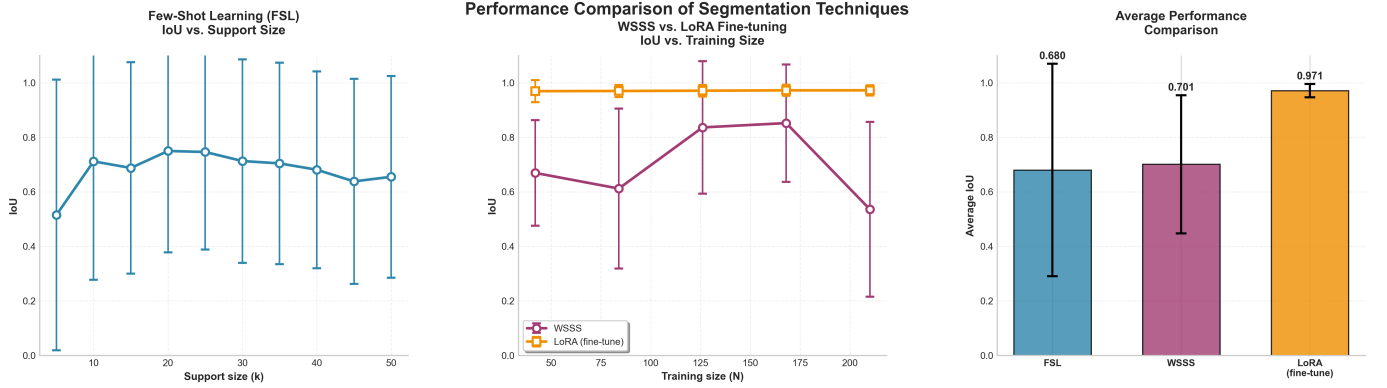


Fig. 1: Comparative analysis of segmentation techniques. (a) FSL performance vs. support size. (b) WSSS vs. LoRA fine-tuning across training sizes. (c) Average performance comparison showing LoRA's superior results (IoU = 0.971). Error bars indicate standard deviation.

TABLE I: Representative public datasets for ophthalmologic segmentation.

Dataset	Modality	Task(s)	Annotation	Size
DRIVE [59]	Fundus	Vessels	Pixel masks (manual)	40 images
STARE [60]	Fundus	Vessels	Pixel masks (2 graders)	20 images
CHASE_DB1 [61]	Fundus	Vessels	Pixel masks (manual)	28 images
HRF [62]	Fundus	Vessels (high-res)	Pixel masks (manual)	45 images
IDRiD [63]	Fundus	DR lesions	Pixel masks (manual)	~81/img per lesion set
REFUGE [64]	Fundus	Optic disc/cup	Pixel masks (manual)	1200 images
ROSE [66]	OCTA	Vessels	Pixel/centerline labels	229 images
RETOUCH [65]	OCT (B-scans)	IRF/SRF/PED	Pixel masks (manual)	90 vols (~11,334 B-scans)
Cataract-1K [67]	Surgical video	Iris/pupil/IOL, tools	Pixel masks (manual)	Frames from 30 videos

b) Segment Anything and derivatives: The *Segment Anything* family advances automatic segmentation via strong generalization and promptable inference. The original **SAM** is a foundation model trained on SA-1B (> 1B masks over 11M images) with a ViT-H encoder and a prompt-conditioned mask decoder (points/boxes/text), enabling zero-shot segmentation without fine-tuning [8]. **SAM2** extends to video with a streaming memory and spatiotemporal attention, trained on VIDEO-SA-1B for consistent real-time segmentation [9]. In medicine, **MedSAM** fine-tunes SAM on > 1.5M image-mask pairs across 10 modalities and 30+ cancer types, improving clinical performance without changing the core architecture [10]; **MedSAM2** adds cross-attention and 2D/3D support (e.g., MRI/CT), trained on 78 public datasets spanning organs such as the eye, lung, brain, and abdomen [11]. For resource-limited settings, **MobileSAM** distills a lightweight encoder (~60× smaller) for efficient CPU/mobile inference with minimal accuracy loss [13], while **FastSAM** reframes the task as instance segmentation with a CNN detector and mask branch, achieving up to 50× SAM's speed using only a fraction of SA-1B and competitive accuracy [12].

D. Public ophthalmologic segmentation datasets

Publicly available datasets for ophthalmologic image segmentation span fundus photography, OCT/OCTA, and anterior-segment surgical video. Canonical resources for vessels (DRIVE, STARE, CHASE_DB1, HRF) provide pixel-wise annotations on small cohorts [59]–[62]. Lesion- and structure-centric sets include IDRiD (DR lesions) [63] and REFUGE (optic disc/cup for glaucoma) [64]. For volumetric imaging, RETOUCH offers OCT fluid masks across 90 volumes [65], while ROSE targets OCTA vessel segmentation [66]. In the anterior segment, CATARACT-1K provides pixel-level masks for key anatomy and instruments in cataract surgery videos [67]. Table I summarizes key characteristics.

Notable gaps in existing datasets include:

- Limited multimodal cohorts (e.g., paired OCT + fundus) and sparse clinical metadata.
- Small sample sizes for rare diseases and advanced stages.
- Variable annotation quality and mask granularity across sources.

III. METHODOLOGY

A. Dataset and context

This study tackles the semantic segmentation of anterior-segment ocular structures under limited supervision. We use 299 images at 224×224 resolution, each paired with a manual mask delineating the pathological region. To examine how data volume affects segmentation quality, we construct five training set sizes for fine-tuning and weakly supervised learning, 42, 84, 126, 168, and 210 images, and ten configurations for few-shot learning with 5, 10, 15, 20, 25, 30, 35, 40, 45, and 50 images. Each configuration is used to train the models, and evaluation is performed on a fixed held-out test set to enable homogeneous comparisons across techniques. Because image selection is random, each experiment is repeated ten times per set size using a fixed seed per run; we report the mean and standard deviation of IoU across runs.

B. Evaluation metric

To assess the quality of the predicted masks we employed common computer-vision metrics that measure both geometric accuracy and perceptual fidelity.

a) Intersection over Union (IoU): This metric quantifies the overlap between a predicted mask and the reference mask. It is defined as the ratio between the area of the intersection and the area of the union of both regions:

$$\text{IoU}(M_{\text{pred}}, M_{\text{ref}}) = \frac{|M_{\text{pred}} \cap M_{\text{ref}}|}{|M_{\text{pred}} \cup M_{\text{ref}}|}, \quad (1)$$

where $|\cdot|$ denotes the number of pixels in the region. This formulation is widely adopted in recent medical segmentation studies [68]. A value of 1 indicates perfect agreement, and a value of 0 means that the regions do not overlap.

C. Fine-tuning a general-purpose segmentation model

The first approach adapts the SAM model, a general-purpose segmentation model, to ocular images. SAM is a transformer-based model capable of segmenting any object given a prompt and was trained on a massive corpus of images and masks, with more than one billion examples [8]. Thanks to this large-scale training, SAM can generalize to new domains without retraining, but it can also be fine-tuned to improve accuracy in specific domains.

To specialize SAM with a small number of images, we applied (LoRA). This technique freezes the model's original parameters and adds low-rank matrices whose multiplication with internal representations corrects the model's behavior. Working with 4-bit quantized weights and memory-efficient optimizers enables fine-tuning with modest computational resources [69]. During training, only the adaptation matrices are optimized while the rest of the model remains unchanged, which helps prevent overfitting. Each training-set size (42, 84, 126, 168, 210) was evaluated in ten independent runs, and the resulting segmentations were compared against the reference masks using the IoU metric.

D. Weakly supervised Learning with Grad-CAM

We begin with a binary cataract/non-cataract classifier built on ResNet-18 [70] initialized from ImageNet. To curb overfitting while retaining informative saliency, only the classifier head and the last convolutional block were unfrozen; all earlier layers remained fixed. After fine-tuning, only images predicted as cataract-positive proceeded to the localization stage. From the last convolutional layer, we extracted Grad-CAM maps, normalized them to $[0, 1]$, and treated them as localization priors. In preliminary trials, *Grad-CAM* produced more stable and reliable maps than classical CAM, so we used it throughout.

Let $f_k(x, y)$ denote the activation of the channel k at spatial location (x, y) in the last convolutional layer and y^c the score for class c . The channel weights and the Grad-CAM map are:

$$\alpha_k^c = \frac{1}{Z} \sum_{x,y} \frac{\partial y^c}{\partial f_k(x, y)}, \quad M_c(x, y) = \text{ReLU}\left(\sum_k \alpha_k^c f_k(x, y)\right).$$

Next, we derived bounding boxes by thresholding each Grad-CAM at a fixed percentile and computing the tight enclosing rectangle with a small margin. These boxes were used to prompt SAM; we also evaluated SAM in its Automatic Mask Generator (AMG) (prompt-free) mode to obtain additional candidate masks. Percentiles for box extraction and mask selection were chosen via small ablations to maximize pseudo-label stability across data splits, without accessing ground-truth masks.

Because no pixel-wise labels are available during training, mask selection is guided by Grad-CAM. Each set of candidate masks returned by SAM is compared against a binarized Grad-CAM support mask (same percentile). The candidate with the highest IoU against this support is adopted as the pseudo-label. This procedure enforces consistency between the localization prior and the foundation model's output while remaining label-free during training.

E. Few-Shot Learning - KDE-based

We cast detection of the *region of interest* as a density-thresholding problem driven by a small positive support set. For each experiment, a support set of size $k \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, \dots, 50\}$ is sampled from the directory of positive (*object*) masks; the query set comprises the remaining positive masks together with all available *background* masks. Instead of training a discriminative classifier, we

estimate the feature distribution of cataract masks with per-coordinate Kernel Density Estimation (KDE) and decide using a log-density gate.

All support and query images are resized to 224×224 , scaled to $[0, 1]$, and normalised with the ImageNet mean and standard deviation. Features $z \in \mathbb{R}^d$ are extracted using a CNN backbone (ResNet-18 [70]), fused into a 768-dimensional embedding.

Given support embeddings $\{z_i\}_{i=1}^k$, we model each coordinate independently with a Gaussian KDE. For dimension j ,

$$\hat{p}_j(u) = \frac{1}{k h} \sum_{i=1}^k \exp\left[-\frac{1}{2} \left(\frac{u - z_{ij}}{h}\right)^2\right], \quad h = k^{-1/5}.$$

Decision thresholds for the aggregate log-likelihood (used at inference) are obtained via the out-of-distribution procedure. Given a small support set of *object of interest* mask embeddings, it yields a similarity or distance score for candidate masks. We estimate the score distribution on the support set and calibrate an acceptance interval $[\theta_{\min}, \theta_{\max}]$. A query mask is deemed *object of interest* if its score falls within this interval; otherwise, it is labeled as *background*. This data-driven calibration adapts the decision boundary to scarce training data without requiring large-scale supervision.

A query image is processed identically to yield z^* . Its log-density score $\ell^* = \sum_j \log \hat{p}_j(z_j^*)$ is compared against the calibrated interval:

$$\hat{y} = \begin{cases} \text{cataract}, & \theta_{\min} \leq \ell^* \leq \theta_{\max}, \\ \text{normal}, & \text{otherwise.} \end{cases}$$

This approach avoids full discriminative training while leveraging the representation power of the fixed backbone.

IV. EXPERIMENTAL DESIGN

A. Splits and Configuration Protocol

For robustness under data scarcity, each experiment is repeated across 10 independent runs per configuration with different random seeds. For methods that require varying training set sizes (fine-tuning and weak supervision), we consider $N \in \{42, 84, 126, 168, 210\}$ images for training and hold out a fixed test set for homogeneous comparison. For few-shot, we evaluate support sizes $k \in \{5, 10, \dots, 50\}$ while keeping the test set fixed across runs. For more information of dataset using in this paper see III-A

B. Treatments

Fine-tuning with LoRA. We adapt SAM [8] using parameter-efficient LORA/QLORA [69]. The base weights remain frozen while low-rank adapters are optimized, enabling domain specialization from small N without overfitting or large compute.

Weakly supervised via classification + GRAD-CAM. A ResNet-18 classifier [70] to detect cataract presence. GRAD-CAM [25] maps from positive images act as localization priors. Among SAM's candidates, the mask with highest IoU versus a binarized GRAD-CAM support map (e.g., $p_{\text{IoU}} \approx 95$) is taken as the pseudo-label (no ground-truth masks accessed during selection).

Few-shot with feature densities. Following prototype-based few-shot ideas [71], we extract fixed embeddings for lesion crops and fit a per-dimension density model on the k support examples. For each query, SAM proposes candidates; we embed each mask region and pick the one with the highest density under the support model, avoiding full discriminative training while leveraging a strong feature extractor.

C. Training Details and Hyperparameters

Unless stated otherwise: batch size 16, Adam with $lr = 10^{-3}$, 3 epochs for classification, and adapter updates. Images are scaled to $[0, 1]$ and normalized with ImageNet statistics. For few-shot, we keep the backbone frozen and vary only the support set. All runs execute on a single NVIDIA L4 GPU. Table II summarizes the configurations used in the experiments.

TABLE II: Summary of the experimental design conducted to evaluate the different paradigms

Component	Configuration	
	Weak-Supervised Learning	Few-Shot Learning (Feature Densities)
Backbone	ResNet-18	
Training setup	3 epochs, batch size 16, Adam ($lr = 10^{-3}$)	
Segmentation models	SAM	
Prompting strategies	AMG	
Mask selection	IoU with Grad-CAM ($p_{IoU} = 95$)	Feature Densities Thresholded
Baseline	Finetuning SAM with LoRA	
Splits	42, 84, 126, 168, 210	5, 10, 15, 20, 25, 30, 35, 40, 45, 50
Metrics	IoU	
Hardware	NVIDIA L4 GPU	

TABLE III: IoU by configuration (mean $\pm$ std) for each technique.

Technique	Config.	IoU (mean $\pm$ std)
FSL	$k = 5$	0.515 ± 0.496
	$k = 10$	0.712 ± 0.434
	$k = 15$	0.688 ± 0.388
	$k = 20$	0.750 ± 0.372
	$k = 25$	0.746 ± 0.357
	$k = 30$	0.713 ± 0.373
	$k = 35$	0.704 ± 0.370
	$k = 40$	0.681 ± 0.361
	$k = 45$	0.638 ± 0.376
	$k = 50$	0.655 ± 0.370
WSSS	$N = 42$	0.670 ± 0.194
	$N = 84$	0.612 ± 0.294
	$N = 126$	0.836 ± 0.243
	$N = 168$	0.852 ± 0.215
	$N = 210$	0.536 ± 0.320
LoRA (fine-tuning)	$N = 42$	0.969 ± 0.040
	$N = 84$	0.970 ± 0.022
	$N = 126$	0.971 ± 0.021
	$N = 168$	0.972 ± 0.020
	$N = 210$	0.972 ± 0.019

V. RESULTS AND DISCUSSION

A. Quantitative results (IoU)

The three approaches (see Table III) behave differently under the *low-data* scenario. **LoRA fine-tuning** on SAM reaches near-ceiling IoU (mean ≈ 0.971) with very little dispersion across configurations (avg. std ≈ 0.025), already from $N = 42$ pixel-wise masks (IoU 0.969) and with marginal gains as N increases to 210. The **WSSS**, which does not use masks during training, achieves mid-level performance (mean ≈ 0.701), peaking at $N = 168$ (IoU 0.852) and declining when percentile calibration drifts away from the optimal region. The **density-based few-shot** variant (FSL) improves with support size and attains its best value at $k = 20$ (IoU 0.750), but exhibits higher variance (avg. std ≈ 0.390).

Overall, the results confirm the *annotation efficiency* of each approach: LoRA capitalizes on a small number of high-quality masks; WSSS turns image-level labels into competitive pseudo-masks; and FSL is a workable option when only a handful of segmented examples can be curated, at the cost of greater run-to-run variability.

B. Aggregate comparison by technique

Aggregating all configurations (Table IV), LoRA leads in average accuracy (IoU ~ 0.971) and stability, making it the preferred choice when a few dozen expert masks are available. WSSS sits in the middle (IoU ~ 0.701), showing that with light calibration (percentiles, margins) image-level labels can be turned into useful segmentations in the absence of pixel-wise annotations. FSL performs well at small k

TABLE IV: Summary by technique across all configurations (mean IoU $\pm$ mean std).

Technique	Avg. IoU $\pm$ Std	Best Config.	Best IoU
FSL	0.680 ± 0.390	$k = 20$	0.750
WSSS	0.701 ± 0.253	$N = 168$	0.852
LoRA (fine-tuning)	0.971 ± 0.025	$N = 210$	0.972

(best at $k = 20$) and minimizes annotation cost, but requires additional strategies (robust support selection, uncertainty filtering) to mitigate variability, especially relevant when each new mask is expensive.

C. Discussion

These findings should be read under the *data scarcity* premise typical of ophthalmic applications. (i) **When pixel-wise masks are available:** LoRA allows a foundation model to be specialized with very few samples and modest compute, reaching near-ceiling IoU; this is the most direct path to deployment if a small, high-quality batch can be annotated. (ii) **When only image-level labels are available:** WSSS offers a practical route to generate pseudo-masks consistent with GRAD-CAM and SAM; its performance depends on light calibration (percentiles/margins). (iii) **When only a handful of masks exist:** FSL reduces curation costs and yields useful results, but its support sensitivity calls for stabilization techniques (active support curation, outlier control and, when possible, averaging across multiple runs).

VI. CONCLUSIONS

This work evaluated three segmentation strategies tailored to *low-data* scenarios in ophthalmology (anterior segment): (i) parameter-efficient fine-tuning of a foundation model (SAM) via LoRA, (ii) WSSS guided by GRAD-CAM and SAM, and (iii) FSL based on feature densities. Using a dataset of 299 images, the main findings are:

- **LoRA** delivers the best accuracy and greatest stability even with few pixel-wise masks: mean IoU close to 0.97 with low dispersion across $N \in \{42, \dots, 210\}$. In resource-limited clinical settings, specializing a foundation model with lightweight adapters is highly effective and computationally economical.
- **WSSS** converts image-level labels into competitive pseudo-masks without requiring dense annotations. It reached mid-to-high IoU (peaking at $N = 168$), though it is sensitive to calibration (percentiles and box margins). It is a practical alternative when expert masks are unavailable.
- **FSL** attains useful IoU with a handful of examples (best at $k = 20$) but shows higher run-to-run variability. Its adoption benefits from robust support selection and uncertainty-control mechanisms to stabilize performance.

In practical terms, we propose a “supervision ladder”: use **LoRA** when at least dozens of masks are available; prefer **WSSS** when only image-level labels exist; and resort to **FSL** when only a few segmented examples can be curated, complementing it with stabilizing strategies (e.g., active support curation, outlier filtering, and averaging across multiple runs).

a) *Limitations and future work.*: The study focused on a single image set and did not incorporate multimodal information; moreover, WSSS depends on GRAD-CAM and fixed thresholds, and FSL is sensitive to support selection. Promising directions include: (i) uncertainty-aware pseudo-label refinement and active instance selection, (ii) cross-domain (device/site) alignment and multimodal data, and (iii) assessing clinical transportability (prospective validation) and lightweight variants for *edge/mobile* deployment.

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